

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.5: AREA OF POLYGONS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we calculate the area of any polygon, no matter how many sides or how irregular, so that we can fairly divide the Kericho community land plot and solve other real Kenyan land and design problems? Answer each section in full sentences. Show all working for calculations, with units. Draw labelled diagrams where they help. Use what you observed and measured in Lessons 1 to 8: the Kericho plot, the Kisumu solar panel, the Mombasa courtyard and the Nakuru school shamba. Give a reason for every claim.

### 1. The Kericho plot: why one formula is not enough

**A village near Kericho has a seven-sided plot to share among three farming families. Nobody at the meeting could find its area with one formula they remembered. (a) Explain why no single standard formula such as the one for a rectangle works for this plot. (b) Describe the method you used to turn the problem into one you can solve. Say how many shapes the plot is split into and how you draw the lines. (c) Explain why the height used in  $A = \frac{1}{2}bh$  must be the perpendicular height and not a slanted side.**

(a) The plot is an irregular heptagon with seven unequal sides and no right angles. It is not a rectangle, square or circle, so no single standard formula fits it. (b) I choose one vertex and draw diagonals from it to every vertex that is not next to it. This splits the heptagon into  $(7 - 2) = 5$  non-overlapping triangles. I measure the base and perpendicular height of each triangle, find each area with  $A = \frac{1}{2}bh$ , and add the five areas:  $A = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \frac{1}{2}b_3h_3 + \frac{1}{2}b_4h_4 + \frac{1}{2}b_5h_5$ . The total of the triangles is the area of the whole plot. (c) The formula comes from a rectangle with the same base and height, and that height is at right angles to the base. A slanted side is longer than the perpendicular height, so using it would overestimate the area of each triangle and of the plot.

### 2. Tools for pieces of the plot: quadrilaterals and the hidden height

**Some pieces of the plot are quadrilaterals, and some triangles have a height that cannot be measured. (a) Write the area formulae you derived**

(a) Parallelogram:  $A = bh$ . Rhombus:  $A = \frac{1}{2}d_1d_2$ . Trapezium:  $A = \frac{1}{2}(a + b)h$ . Kite:  $A = \frac{1}{2}d_1d_2$ . If I cut a triangle off one slanted end of the parallelogram and move it to the other end, I get a rectangle with the same base  $b$  and perpendicular height  $h$ , so  $A = bh$ . The rhombus and kite share a formula because each is fixed by its two diagonals. (b) The panel is mounted at an angle on the roof, so its perpendicular height cannot be measured directly. Only two sides and the angle between them are known. (c) Drop a perpendicular from a vertex to side  $b$ . In the right-angled triangle formed,  $\sin C = h \div a$ , so  $h = a \sin C$ . Substituting into  $A = \frac{1}{2}bh$

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| <p>for the parallelogram, rhombus, trapezium and kite. For the parallelogram, explain how cutting and rearranging it shows its formula. (b) In Kisumu, a triangular solar panel has two sides of 5 m and 7 m with an included angle of <math>58^\circ</math>. Explain why <math>A = \frac{1}{2}bh</math> is hard to use here. (c) Derive the formula <math>A = \frac{1}{2}ab \sin C</math> from <math>A = \frac{1}{2}bh</math>, then calculate the area of the panel. Give your answer to 2 decimal places.</p> | <p>gives <math>A = \frac{1}{2}b(a \sin C) = \frac{1}{2}ab \sin C</math>. For the panel: <math>A = \frac{1}{2} \times 5 \times 7 \times \sin 58^\circ = 17.5 \times 0.8480 = 14.84 \text{ m}^2</math> (calculator and mathematical tables give the same value). So every triangle in the Kericho plot can be solved from two sides and the included angle, even when its height is hidden.</p> |
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### 3. Checking with a regular polygon and a scale diagram

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| <p>(a) The Mombasa Old Town courtyard is a regular octagon with side 3 m. Explain how drawing lines from the centre O to each vertex lets you find its area, and state the central angle of each triangle. (b) Calculate the area using the direct formula <math>A = 2(1 + \sqrt{2})s^2</math>, to 2 decimal places. (c) Explain why the formula and the decomposition method must agree. (d) On the Nakuru school shamba diagram, <math>1 \text{ cm}^2</math> represents <math>16 \text{ m}^2</math>. One triangle on the diagram has an area of <math>3 \text{ cm}^2</math>. Find its real area and explain why the conversion for area uses 16 rather than the conversion factor for lengths.</p> | <p>(a) The lines from O split the octagon into 8 congruent isosceles triangles. Each central angle is <math>360^\circ \div 8 = 45^\circ</math>. The area of one triangle is <math>\frac{1}{2} \times \text{base} \times \text{height}</math>, where the base is the side (3 m) and the height is the apothem. The octagon's area is 8 times one triangle. (b) <math>A = 2(1 + \sqrt{2}) \times 3^2 = 2 \times 2.4142 \times 9 = 43.46 \text{ m}^2</math>. (c) The formula comes from the decomposition: <math>8 \times</math> the area of one isosceles triangle simplifies to <math>2(1 + \sqrt{2})s^2</math>. It is not a separate rule, so the two methods give the same area, apart from small measurement errors. (d) Real area = <math>3 \times 16 = 48 \text{ m}^2</math>. Area is measured in square units, so the linear scale factor is squared: if 1 cm represents k metres, then <math>1 \text{ cm}^2</math> represents <math>k^2</math> square metres. Here <math>k^2 = 16</math>. Using the length factor alone would convert a length, not an area.</p> |
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### 4. Dividing the Kericho plot fairly

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| <p>The Chepkemoi, Mwangi and Otieno families each need an equal area of the Kericho plot. (a) Explain how you would find</p> | <p>(a) I split the plot into triangles and quadrilaterals whose areas I can find with known formulae, then add them: <math>\text{Area}(\text{composite}) = \sum \text{Area}(\text{components})</math>. If a shape is a larger polygon with a section removed, I find the whole area and subtract the removed part: <math>\text{Area}(\text{composite}) = \text{whole} - \text{removed section}</math>. (b) Each family must receive one-third of the total, so each share is <math>\text{total area} \div 3</math>. I then draw boundary lines so that the land on each side adds up to exactly that area, and check each share by calculating its area.</p> |
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| <p>the total area of the plot. Describe how composite shapes can be handled with both adding and subtracting. (b) Explain how the three families' shares are worked out from the total. (c) Two groups split the plot in different ways and get totals that differ slightly. Give two possible reasons for the difference. State what it would mean if their methods were both correct and carried out accurately.</p> | <p>(c) Possible reasons: measurement errors with the ruler, or a wrong perpendicular height, such as using a slanted side. Other reasons are overlapping triangles, missing a region, or a mistake converting <math>\text{cm}^2</math> to <math>\text{m}^2</math>. If both methods are correct and accurate, the totals must be equivalent, because the area of a polygon does not depend on how it is divided into parts. This is why different valid decompositions give the same area.</p> |
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| 5. Answering the driving question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <p>Write a short final answer to the driving question. (a) Explain how you can calculate the area of any polygon, no matter how many sides or how irregular. Include how many triangles an n-sided polygon is split into from one vertex. (b) Give one real Kenyan situation from our lessons where polygon area is used and explain why an accurate area matters in it. (c) Compare your first estimate of the Kericho plot in Lesson 1 with how you now solve the problem. Describe what you now understand better.</p> | <p>(a) Any polygon can be divided into non-overlapping triangles whose areas I can calculate, and the polygon's area is the sum of those areas. From one vertex, an n-sided polygon splits into <math>(n - 2)</math> triangles, for example 5 for the 7-sided Kericho plot and 6 for an octagon. Where the shapes are standard I can use the quadrilateral formulae directly. If a height cannot be measured, I use <math>A = \frac{1}{2}ab \sin C</math>. For a scale diagram, I multiply by the squared scale factor to get real areas. (b) In Kericho, an accurate area decides how tea and maize land is shared fairly among three families, which affects their food and income. Other examples are the Mombasa courtyard (the number and cost of tiles) and the Kisumu solar panels (how many panels are needed). (c) In Lesson 1 my estimate was only a visual guess. Now I can measure and calculate the area step by step, check it with a second method, and explain any difference from the guess. I understand that the triangle is the building block for the area of every polygon.</p> |

| RUBRIC                                               |                                                                                                                                                                                             |                                                                                                                                      |                                                                                                        |
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| Criterion                                            | Excellent (4)                                                                                                                                                                               | Proficient (3)                                                                                                                       | Developing (1–2)                                                                                       |
| Decomposition method and the $(n - 2)$ triangle idea | Clearly explains drawing diagonals from one vertex, states $(n - 2)$ triangles correctly for the heptagon and other polygons, and explains that the total is the sum of the triangle areas. | Describes splitting into triangles and summing areas, with the number of triangles mostly correct or the reasoning partly explained. | Mentions shapes but does not explain how the polygon is split or how the parts combine into the total. |

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| Use of formulae and calculation accuracy                             | Recalls all formulae correctly, derives $A = \frac{1}{2}ab \sin C$ , and calculates $14.84 \text{ m}^2$ , $43.46 \text{ m}^2$ and $48 \text{ m}^2$ with correct working and units.                 | Uses most formulae correctly and gets most calculations right, with minor slips in rounding or units. | Formulae are missing or confused (for example slant height instead of perpendicular height) and calculations are incomplete or incorrect. |
| Reasoning and links between methods                                  | Explains why the perpendicular height is needed, why the octagon formula and decomposition agree, why area scale factors are squared, and why different valid decompositions give the same area.   | Gives reasons for most of these ideas, though some explanations are brief or incomplete.              | States answers without reasons or does not connect methods to each other.                                                                 |
| Application to the Kericho land-division problem and Kenyan contexts | Clearly explains finding the total area, dividing it into three equal shares for the families, handling composite shapes by adding and subtracting, and the real-life importance of accurate area. | Explains how the plot is divided and gives a relevant Kenyan example, with some gaps in detail.       | Gives a vague or incomplete explanation of the land division and little or no link to real contexts.                                      |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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